



PEOPLE'S EDUCATION TRUST (R), MANDYA

P.E.S. COLLEGE OF ENGINEERING MANDYA

(An Autonomous Institution under VTU, Belagavi, Aided by Govt. of Karnataka.)



Project Report on

“An Agentic Business to Business (B2B) Marketplace for the Automation and Simplification of Global Trade for Small and Medium-sized Enterprises (SMEs)”

Submitted in partial fulfilment of the requirements
for the award of the Degree

BACHELOR OF ENGINEERING

in

Computer Science and Engineering (AI & ML)



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This is to certify that the work entitled “**An Agentic B2B Marketplace for the Automation and Simplification of Global Trade for SMEs**” is a bona fide work carried out by **Komal N (4PS22CI060)**, A Student of VII Semester, Bachelor of Engineering in Computer Science and Engineering (AI & ML) at PES College of Engineering, Mandya, under Vishveshvaraya Technological University (VTU), Belagavi, during the academic year 2025–2026. The project report has been approved as it satisfies the academic requirements prescribed for the Project work of VII Semester.

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DECLARATION

We hereby declare that the project titled “**An Agentic B2B Marketplace for the Automation and Simplification of Global Trade for SMEs**” is carried out by Komal.N (4PS22CI060), under the guidance of Prof. Ashwini M.C, Assistant Professor, Department of Computer Science and Engineering (AI & ML), PES College of Engineering, Mandya, in partial fulfilment of the academic requirements for the Mini Project work of VI Semester, Bachelor of Engineering under Visvesvaraya Technological University (VTU), Belagavi, during the academic year 2025–2026.

We also declare that this project has not been submitted to any other university or institution for the award of any degree or diploma.

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ABSTRACT

Global supply chains are undergoing rapid digital transformation under the paradigm of Logistics 4.0. Despite this progress, **Small and Medium-sized Enterprises (SMEs)** continue to face significant challenges in accessing and participating in international trade. Regulatory complexity, extensive documentation requirements, limited access to trade finance, and fragmented digital systems increase operational friction and reduce competitiveness for SMEs in global markets.

This project presents **TradeOS**, an agentic, AI-driven B2B marketplace designed to automate, streamline, and secure the complete trade lifecycle for SMEs. The proposed system is structured as a **tri-layered architecture** consisting of: (1) a **Presentation Layer** enabling verified partner discovery, (2) an **Intelligence Layer** incorporating predictive analytics and forecasting models, and (3) an **Automation Layer** that manages end-to-end trade execution.

At the core of TradeOS is an **Agentic AI Co-pilot**, powered by a **Multi-Agent System (MAS)** capable of coordinating trade processes autonomously. The system integrates specialized agents such as the **DocuAgent**, which employs Transformer-based NLP for automated documentation analysis and HS Code classification; the **RiskAgent**, which performs multi-factor, real-time risk assessment using stochastic modelling; the **LogisticsAgent**, which computes optimal logistics routes; and the **EscrowAgent**, which ensures secure, milestone-based payment settlement through smart contracts on a **Distributed Ledger Technology (DLT) Trust Layer**.

The project follows the **Agile Scrum methodology**, with iterative development and continuous validation. A pilot test route (Bengaluru → Hamburg) is used to evaluate system performance.

Overall, TradeOS aims to provide SMEs with a unified, intelligent, and secure platform for global trade, enhancing supply chain resilience and setting a new benchmark for transparent and user-centric trade automation.

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Chapter 1: Introduction

1.1 Introduction

Small and Medium-sized Enterprises (SMEs) are critical drivers of global trade, yet they continue to face barriers such as regulatory complexity, lack of reliable partners, financial risks, and limited supply chain visibility. These challenges restrict their ability to compete with larger corporations in international markets. Recent advancements in Artificial Intelligence, blockchain and multi-agent systems offer an opportunity to simplify and automate the trade process. However, most digital trade platforms are designed for large enterprises, leaving SMEs underserved.

This project, **TradeOS**, proposes an agentic B2B marketplace that integrates a cognitive co-pilot powered by multi-agent systems to manage the entire trade lifecycle—from compliance and documentation to logistics and payment settlement. The aim is to democratize global trade by reducing operational friction and enabling SMEs to participate more efficiently in international markets

1.2 Problem Statement

The participation of Small and Medium-sized Enterprises (SMEs) in the global value chain is critically restricted by **information asymmetry** and **operational complexities**. Despite their significant contribution to the economy, SMEs face challenges that prevent them from fully engaging in international trade. The major issues addressed in this research are as follows:

- SMEs struggle to interpret and comply with diverse international trade regulations, tariffs, and customs documentation standards, leading to costly delays and penalties
- SMEs are highly vulnerable to financial risks such as currency fluctuations and lack access to advanced trade finance instruments
- Lack of end-to-end supply chain visibility exposes SMEs to disruptions caused by geopolitical instability, extreme weather, and logistical bottlenecks.

The core problem is the absence of an intelligent, automated, and collaborative platform that integrates partner discovery, regulatory compliance, risk assessment, and financial settlement into a unified workflow, specifically for SMEs

1.3 Core Objectives

The main objectives of the project are:

- To design a **tri-layered system architecture** (Marketplace, Intelligence, Automation) for a scalable and secure B2B trade platform.
- To architect and validate a **cognitive multi-agent system** (MAS), potentially employing a Belief-Desire-Intention (BDI) model, for autonomous trade execution.
- To engineer a suite of specialized agents, including a **DocuAgent** utilizing a hybrid OCR and Transformer-based NLP model for semantic data extraction, and a **RiskAgent** employing stochastic modeling for multi-factor risk assessment.
- To develop and test a prototype on a defined trade route (e.g., Bengaluru, India to Hamburg, Germany) to quantify the reduction in process lead time and compliance errors.

1.4 Scope

The project focuses on developing a functional prototype of the TradeOS platform. The scope includes:

- **Architecture:** Designing the three operational layers—Presentation, Intelligence, and Automation—along with a DLT-based Trust Layer.
- **Agent Development:** Implementing core logic for five key agents: MatchmakerAgent, DocuAgent, RiskAgent, LogisticsAgent, and EscrowAgent.
- **Validation:** Demonstrating the complete trade process through a sample trade scenario.
- **Technology Stack:** Using Python, FastAPI, and agent-frameworks such as LangChain or Autogen for system development.

To conclude, this chapter establishes the foundation for the TradeOS project by outlining the challenges faced by SMEs, the core motivations, and the objectives driving the proposed solution. The definitions of scope and system goals presented here serve as the basis for the architectural design and technical methodology discussed in the subsequent chapters.

Chapter 2: Literature Survey

2.1 Existing Work and Research Gaps

Existing literature highlights significant advancements in individual components of global trade automation, such as AI-based HS code classification, blockchain-enabled traceability, and multi-agent systems for supply chain decision-making. However, these solutions remain fragmented and are not designed specifically for SMEs.

Key gaps identified include:

- Lack of integration across compliance, logistics, risk modelling, and payments.
- Limited automation, resulting in continued dependency on manual workflows.
- High complexity and cost of existing ERP/SCM platforms, making them unsuitable for SMEs.
- Absence of an end-to-end intelligent system that can autonomously execute complete trade workflows.

2.2 Analysis of Existing Systems

Current B2B trade systems suffer from operational fragmentation and limited intelligence:

a) Manual Brokerage Processes

SMEs commonly rely on emails, spreadsheets, and paper-based coordination for freight forwarding, customs clearance, and payments. These disconnected practices lead to frequent errors, poor visibility, and heavy reliance on intermediaries.

b) Digital Marketplaces

Platforms like Alibaba.com provide global supplier discovery but lack automated compliance verification, logistics planning, and risk assessment. Execution remains largely manual.

c) ERP and SCM Platforms

Enterprise-grade solutions such as SAP Ariba and Oracle SCM offer integrated workflows but are costly, resource-intensive, and too complex for SME adoption.

d) Specialized Digital Tools

Standalone tools exist for HS code classification (NLP-based), blockchain trade finance, and multi-agent logistics research. However, they operate independently and lack interoperability, forcing SMEs to manually manage multiple systems.

Summary:

No existing system provides SMEs with a unified, intelligent, and fully automated trade execution environment. This highlights the need for an integrated agentic platform like TradeOS.

2.3 Limitations of Existing System

- **Lack of Integration Across Functions**

Current systems operate in silos, where key processes such as customs **Lack** documentation, logistics management, and payment handling are not integrated into a unified platform. This fragmentation forces SMEs to rely on multiple tools and platforms, resulting in inefficiencies, increased operational complexity, and higher chances of data inconsistency.

- **Not Tailored for Small and Medium Enterprises (SMEs)**

Most existing solutions are designed to meet the needs of large enterprises, making them expensive and technically complex for SMEs. Small businesses often lack the financial and technical resources to adopt such systems, limiting their participation in global trade.

- **Limited Automation and High Manual Dependency**

Although partial automation exists, many critical processes still require manual intervention. Tasks such as customs documentation, compliance verification, and transaction processing are often performed manually, leading to delays, human errors, and reduced productivity.

- **Lack of Trust and Transparency**

Existing systems do not provide a comprehensive mechanism for verifying the authenticity and credibility of trading partners. This lack of transparency increases the risk of fraud and disputes, especially in cross-border transactions involving unfamiliar entities.

2.4 Literature Review Summary

Author(s)	Title	Dataset/Source	Methodology	Relevance to TradeOS
Chen et al.	Neural Machine Translation for HS Code Prediction	1.156M DHL shipment records	Attention-based NMT	Forms basis for DocuAgent's HS Code automation
Xu et al.	Will Bots Take Over the Supply Chain?	132 papers (1993–2020)	Systematic Literature Review	Supports MAS architecture for automation
Chang & Chen	Blockchain in Supply Chain	106 articles	SLR	Guides Trust Layer and EscrowAgent design
WTO (2023)	World Trade Report	WTO & World Bank data	Economic modelling	Validates digital trade opportunity for SMEs
WTO (2016)	SME Trade Barriers Report	Global SME survey data	Policy & economic analysis	Defines SME-specific obstacles addressed by TradeOS
Forget et al.	Multi-Behaviour Agent Model	Industry experimental platform	Hybrid agent modelling	Informs RiskAgent and LogisticsAgent behaviour design

2.5 Conclusion

The literature clearly shows that although individual technologies—AI, MAS, blockchain—have matured, they remain siloed and inaccessible to SMEs. There is no unified system that integrates compliance, risk, logistics, and payments into a single intelligent workflow. This gap directly motivates the development of TradeOS, an agentic platform tailored to simplify and automate global trade for SMEs.

Chapter 3: Proposed System

3.1 Overview of the Proposed System

TradeOS is a **Smart Agentic B2B Marketplace** designed to enable Small and Medium-sized Enterprises (SMEs) to participate efficiently and confidently in global trade. The platform directly connects verified international buyers and sellers, removing intermediaries and improving transparency, trust, and operational control. The system integrates an **Agentic AI Engine** that automates complex trade tasks such as documentation, HS Code classification, compliance checks, logistics planning, risk evaluation, and secure payment execution. By replacing manual workflows with autonomous software agents, TradeOS establishes a reliable, efficient, and intelligence-driven trade ecosystem that enhances SME competitiveness.

Technically, TradeOS consolidates fragmented trade operations into a **coordinated multi-agent execution framework**. Each phase of the trade lifecycle—partner discovery, document generation, risk modelling, logistics routing, and settlement—is handled by agents that collaborate and make context-aware decisions in real time.

3.2 Key Features

The TradeOS platform incorporates several advanced features to enable seamless, secure, and intelligent trade execution for SMEs:

- **Smart B2B Marketplace** : A digital platform that connects verified buyers and sellers globally, eliminating intermediaries and improving trust and transparency.
- **Agentic AI Automation** : The system employs multiple intelligent agents to automate critical trade processes such as partner discovery, documentation, risk analysis, logistics planning, and payment handling.
- **HS Code Classification and Document Generation** : Automated classification of goods using AI models and generation of customs-compliant documents, reducing manual effort and errors.
- **Risk Analysis and Predictive Modelling** : Real-time evaluation of risks based on external data sources such as weather conditions, port congestion, and market fluctuations.

- **Optimized Logistics Planning** : Intelligent route selection and shipment planning using APIs and optimization algorithms to minimize cost and delivery time.
- **End-to-End Integration** : All stages of the trade lifecycle are integrated into a single platform, improving efficiency and reducing dependency on multiple systems.

3.3 Proposed System Workflow

The operational workflow of TradeOS is executed through a coordinated set of intelligent agents. These agents communicate asynchronously, enabling scalable automation and fault tolerance.

Workflow Steps:

Step 1: SME Onboarding

The platform performs digital KYC and verification of trade credentials using rule-based logic and API-supported authentication, ensuring that only legitimate businesses access the marketplace.

Step 2: Trade Partner Discovery

The **MatchmakerAgent** identifies optimal partners using similarity scoring, product taxonomy filters, and historical interaction heuristics. This reduces information asymmetry and accelerates partner selection.

Step 3: Deal Initiation

Once trade terms are agreed upon, the **Agentic Co-pilot** activates and begins orchestrating all downstream operations.

Step 4: Autonomous Multi-Agent Execution

A coordinated set of intelligent agents handles mission-critical tasks:

- **DocuAgent**: Performs HS Code classification using transformer-based NLP and generates structured customs documents.
- **RiskAgent**: Conducts risk estimation using real-time data (weather, port congestion, currency volatility).
- **LogisticsAgent**: Computes optimal shipping routes by interfacing with logistics APIs and applying risk constraints.
- **EscrowAgent**: Executes secure, milestone-based payments via blockchain smart contracts.

Agents communicate asynchronously to ensure modular, scalable, and fault-tolerant execution.

Step 5: Completion and Settlement

Upon delivery confirmation, the EscrowAgent verifies the final state and releases payment automatically through smart contract logic. This ensures immutability, transparency, and dispute-free settlement.

3.4 Trade Lifecycle Automation Framework

TradeOS automates every phase of cross-border trade:

1. **Verification Layer** – Validates SMEs before entry.
2. **Discovery Layer** – Identifies compatible partners algorithmically.
3. **Orchestration Layer** – Activates the Agentic Co-pilot to execute tasks.
4. **Execution Layer** – Agents perform documentation, risk analysis, routing, and payment control.
5. **Settlement Layer** – Smart contracts finalize payments securely.

This framework ensures deterministic automation, reducing errors and manual bottlenecks.

3.5 Feasibility Study

A. Technical Feasibility

The system is technically feasible due to the availability of:

- LangChain/Autogen for agent-based automation
- FastAPI for backend integration
- Python-based NLP and ML libraries
- Blockchain systems for secure payment execution

B. Operational Feasibility

The modular multi-agent architecture allows independent development, testing, and scaling of components, ensuring maintainability.

Chapter 4: Methodology

The methodology for the proposed system, **TradeOS**, follows a structured multi-stage pipeline designed to automate international trade workflows using a multi-agent architecture. The system integrates supplier discovery, risk evaluation, document generation, and logistics planning into a unified framework. The process ensures scalability, accuracy, and real-time decision-making.

4.1 Workflow and Stages

The TradeOS system operates through the following stages:

Stage 1: Data Collection

Sources:

Data is aggregated from multiple structured and real-time sources, including:

- International trade datasets (HS code-based trade data)
- Supplier datasets generated from Kaggle
- Government trade portals
- Commodity price datasets
- Logistics and routing datasets

APIs and Tools:

- Market price APIs (data.gov.in, commodity APIs)
- Weather APIs (OpenWeather)
- Currency exchange APIs (ExchangeRate API)
- Route APIs (OpenStreetMap / OSRM)

Volume:

- Handles large-scale structured datasets (1000–2000+ records)
- Supports semi-real-time updates for dynamic risk evaluation

Stage 2: Data Pre-processing

Data Cleaning:

- Removal of missing and inconsistent data
- Standardization of HS codes and country names
- Normalization of numerical attributes

Feature Engineering:

- Derived fields such as:
 - reliability_score
 - volatility_index
 - trade_frequency
 - logistics_distance

Noise Filtering:

- Removal of unrealistic synthetic entries
- Filtering irrelevant commodities or inactive trade routes

Stage 3: Matchmaker Agent (Supplier-Buyer Matching)**Matching Logic:**

- HS Code similarity
- Country compatibility
- Supplier reliability score
- Capacity vs demand alignment

Algorithms Used:

- Rule-based filtering
- Weighted scoring model

Output:

- Top N supplier recommendations
- Ranked based on:
 - success rate
 - cost efficiency
 - historical trade performance

Stage 4: Risk Assessment Agent**Risk Factors Analyzed:**

- Supplier risk (reliability, disputes)
- Financial risk (currency volatility)
- Logistics risk (route delay, weather)
- Compliance risk (tariffs, restrictions)
- Market risk (price fluctuations)

Data Sources:

- Weather API (OpenWeather)
- Currency API (ExchangeRate-API)
- Trade datasets (UN Comtrade)

Scoring Mechanism:

- Weighted aggregation model:
$$\text{Risk Score} = f(\text{Supplier} + \text{Financial} + \text{Logistics} + \text{Compliance} + \text{Market})$$

Output:

- Risk score (0–10)
- Risk classification (Low / Medium / High)
- Recommendations (Escrow, alternate route, supplier change)

Stage 5: Document Agent**Functionality:**

- Generates trade-compliant documents:
 - Sales Contract
 - Invoice
 - Letter of Credit draft
 - Bill of Lading draft

Approach:

- Template-based generation
- Rule-based compliance checks

Validation:

- Ensures:
 - HS code correctness
 - Trade term consistency
 - Mandatory field completeness

Stage 6: Logistics Agent**Core Tasks:**

- Route optimization
- Mode selection (Sea / Air / Land)

- Carrier selection
- Shipment planning

APIs Used:

- Routing: OpenStreetMap
- Weather: OpenWeather

Decision Logic:

- Mode selection based on:
 - delivery deadline
 - quantity
 - risk score

Output:

- Optimal route
- Estimated cost and delivery time
- Selected logistics carrier

Stage 7: Execution & Tracking

Shipment Monitoring:

- Tracks shipment stages:
 - Dispatched
 - In transit
 - Delivered
 - Delayed

Event Detection:

- Identifies:
 - port delays
 - weather disruptions
 - route deviations

Stage 8: Visualization & User Interface

Dashboard Features:

- Supplier recommendations
- Risk analysis reports
- Logistics tracking
- Trade insights

Tools:

- Interactive dashboards (e.g., Power BI / web UI)
- API-driven frontend

Stage 9: Evaluation and Validation**Performance Metrics:****1. Matching Accuracy:**

- Precision of supplier recommendations

2. Risk Model Evaluation:

- Consistency of risk scoring

3. Logistics Optimization:

- Cost vs time efficiency

4. Validation Techniques:

- Cross-check with historical trade patterns
- Scenario-based simulation testing

5. Benchmarking:

- Compared against traditional trade workflows
- Evaluates automation efficiency and decision accuracy

Chapter 5: System Design

5.1 System Architecture

TradeOS is designed using a **three-layer distributed architecture** supported by an **Agentic Intelligence Layer** and a **Multi-Agent Automation Engine**. This separation of concerns ensures efficient processing, secure data handling, and robust multi-agent coordination.

Tri-Layer Architecture

1. Presentation Layer (User Interaction Interface)

- Built using **React.js** and API-driven communication.
- Handles SME onboarding, KYC verification, partner discovery, trade initiation, and tracking dashboards.
- Communication occurs through an API gateway for secure access control.

2. Intelligence Layer (Cognitive Decision Core)

- Hosts AI-driven components:
 - Transformer-based HS Code classification
 - NLP-driven document parsing
 - Stochastic risk prediction models
 - Market and logistics analytics
- Provides inference outputs to the Automation Layer.
- Enables reasoning, context sharing, and dynamic decision-making.

3. Automation Layer (Agentic Co-Pilot)

- Implements autonomous trade operations using a **Multi-Agent System (MAS)** built with LangChain/Autogen.
- Each agent executes domain-specific tasks and communicates through an event-driven messaging bus (Kafka/RabbitMQ).
- Ensures deterministic, rule-driven execution of trade workflows.

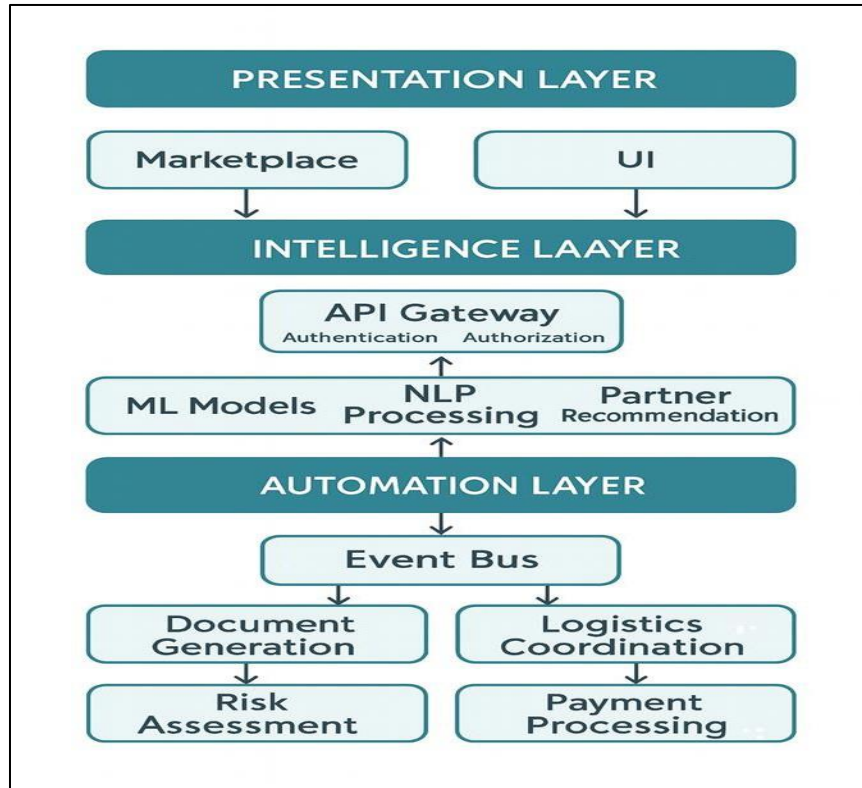


Fig 4.1 Three-layer distributed architecture

Agentic Intelligence Layer

This layer enables coordinated decision-making across all autonomous agents. It provides:

- A shared **knowledge base** (market data, partner profiles, risk datasets)
- Workflow context propagation
- FIPA-style ACL message formats
- An event-driven communication hub for inter-agent collaboration

Specialized Agents in TradeOS

Agent	Technical Role
MatchmakerAgent	Identifies compatible SMEs using similarity scoring and product taxonomy.
DocuAgent	Performs OCR, HS Code classification, semantic document parsing, and document generation.
RiskAgent	Evaluates logistics and financial risks using stochastic models and real-time data feeds.
LogisticsAgent	Computes optimal shipping routes using carrier APIs and risk constraints.
EscrowAgent	Executes secure smart-contract payments and milestone-based fund release.

High-Level System Architecture

1. Presentation Layer (Marketplace & UI) :

Web and mobile frontends for user onboarding, partner discovery, trade initiation, dashboards, and document review. Exposes REST/GraphQL APIs to the backend.

2. Platform / Orchestration Layer :

API gateway, authentication & authorization (OAuth2 / JWT), request routing, and service mesh for secure inter-service communication.

3. Intelligence Layer (Cognitive Core) :

ML/NLP services for HS-code classification, demand forecasting, risk scoring, and partner recommendation. Model hosting (TF/PyTorch + model registry) with batch and real-time inference endpoints.

4. Automation Layer (Agentic Co-pilot) :

Multi-agent system (MatchmakerAgent, DocuAgent, RiskAgent, LogisticsAgent, EscrowAgent) that coordinates trade workflows, issues tasks, and executes automated actions via well-defined agent APIs.

5. Data & Integration Layer :

Event bus / message queue (Kafka or RabbitMQ) for asynchronous workflows, ETL pipelines, and microservices. Connectors for external systems: customs databases, carrier APIs, market data feeds, and payment gateways.

6. Trust & Persistence Layer:

Transactional database (Postgres) for core records, time-series store for telemetry, object store for documents, and a distributed ledger / blockchain module for tamper-proof audit trails and escrow smart contracts.

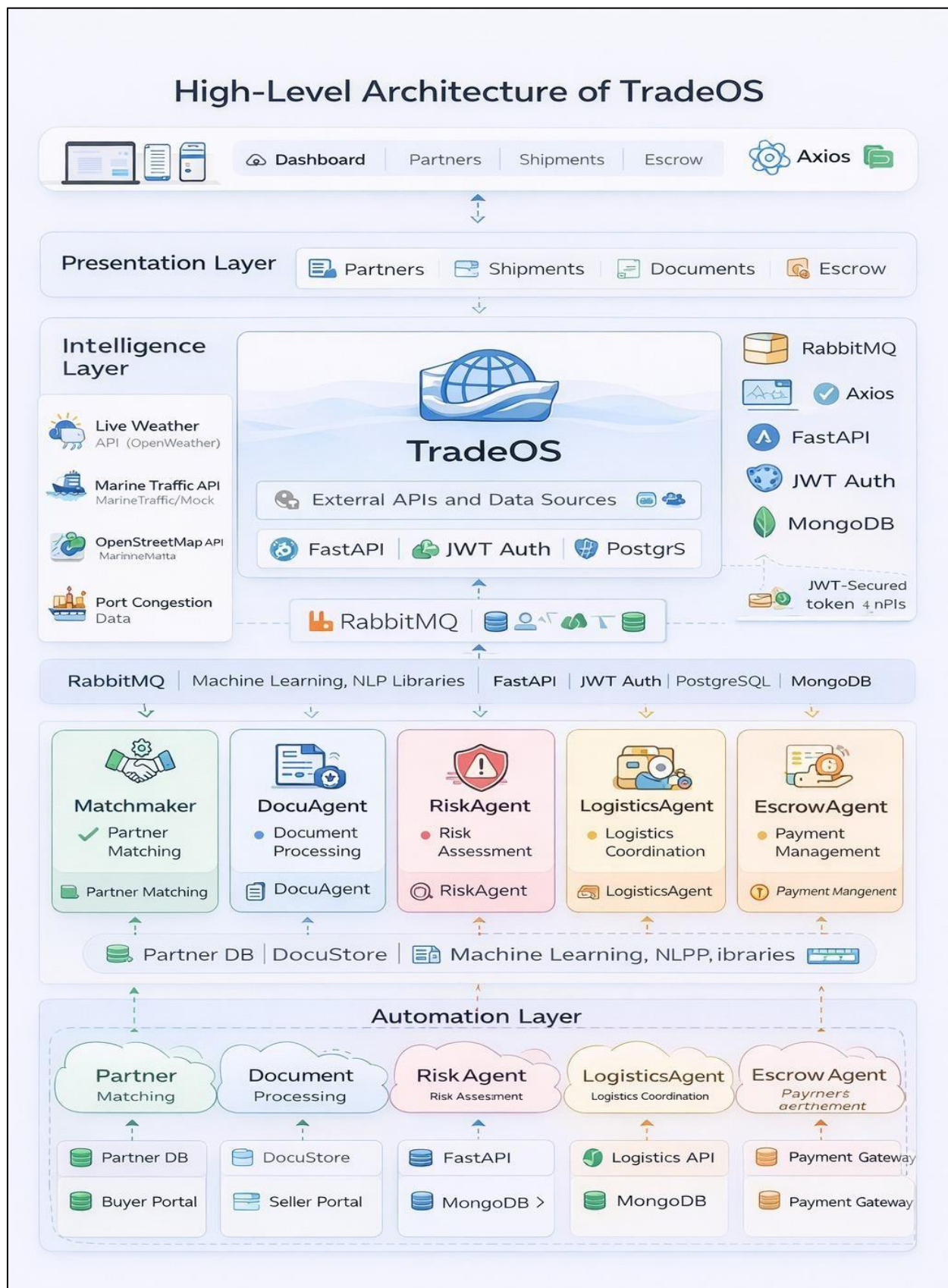
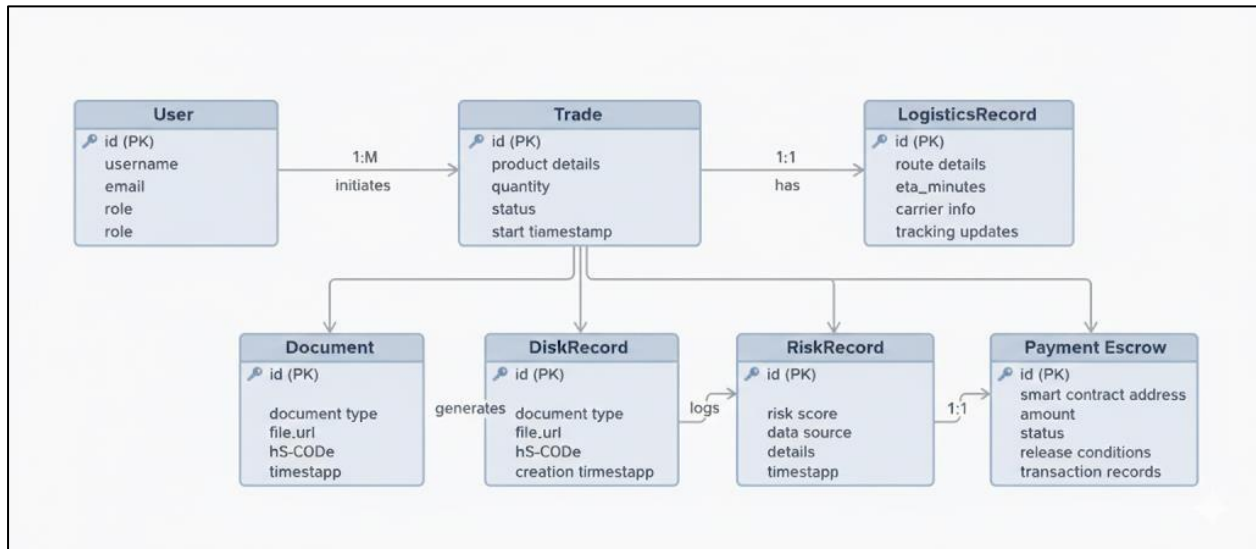


Fig 4.2 High-Level Architectur

5.2 Database Design

The system uses a relational database (PostgreSQL) for secure, normalized storage of user, trade, logistics, and compliance data.

5.2.1 ER Diagram



5.2.2 Database Schema

The following SQL Data Definition Language (DDL) statements define the structure of the three main tables in the database.

```
<> SQL Schema: Trade Platform Database

-- User Table: Stores of profiles
CREATE TABLE users (
    id TEXT PRIMARY KEY,
    username TEXT UNIQUE NOT NULL,
    email TEXT UNIQUE NOT NULL,
    password_hash TEXT,
    role TEXT DEFAULT 'buyer',
);

-- e.g. insert new users
INSERT INTO users (id, username, email, password_hash, role) VALUES ('new_process', 'new_process', 'new_process@users.com', 'new_process', 'new_process');

-- Resources Tables
CREATE TABLE IF NOT EXISTS trades (
    id TEXT PRIMARY KEY,
    type TEXT NOT NULL,
    latitude REAL NOT NULL,
    longitude REAL NOT NULL,
    status TEXT NOT NULL,
);

-- Assignments Table: Links an incident to a resource and stores the plan
CREATE TABLE IF NOT EXISTS assignments (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    incident_id TEXT NOT NULL,
    resource_id TEXT NOT NULL,
    route_instructions TEXT,
    eta_minutes INTEGER,
    status TEXT NOT NULL DEFAULT 'pending',
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
);
```

5.2.3 Data Dictionary

The following tables provide a detailed description of each column within the database schema.

Table 1.1.1.1: LogisticsRecord							
Field		Description	Data Type			Denstrations	
id	route	Selectics path	TEXT	PRIMARY KEY		PRIMARY KEY	NOT NULL
saroe	carrier_api	estimator used	TEXT	carrier used		NOT NULL	TEXT
factors	estimated delivery	JSN array of	REAL	NOT TIME		DATETIME	The acking of tracking ray date
tracking updates	timestamp	tracking events	JONB	DATETIME		DATETIME	

Table 1.1.1. 2: RiskRecord							
Field			Data Type			Destrations	
id	Primary key	Foreign text	trade_id	PRIMARY KEY	Blockchain smart address	e.g.I., "pending, ", released, (d, ,rlesassed",	
trade	trade_id	riskore (0-100)	INTEGER	TEXT, FORIGIN KEY	contract ,reblited", disputed	contucunding, TER, NOT TX)	
amount	NUMERIC	JSN object of	NEXT	FOREAN KEY	TEXT	JON any of bloclnain TXs	
release conditions	NUMERIC	risk assesments	NEXT	DATETIME		JSNB	

Table 1.1.1.3: PaymentEscoro							
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5.3 Workflow and Data Flow Diagrams

5.3.1 DFD Level 0 (System Context)

External Entities: Buyer SME, Seller SME

Central Process: TradeOS

Data Stores: Users DB, Trade DB, Logistics DB

Flows:

5.3.1.1 SMEs → registration, trade data → TradeOS

5.3.1.2 TradeOS → risk, routing, documents → Databases

5.3.1.3 TradeOS → updates → SMEs

5.3.2 DFD Level 1 (Detailed Breakdown)

Modules:

1. User Verification
2. MatchmakerAgent (Partner Discovery)
3. DocuAgent (Document Processing)
4. RiskAgent (Risk Evaluation)
5. LogisticsAgent (Route Planning)
6. EscrowAgent (Payment Settlement)

Each module interacts with its corresponding database table and triggers subsequent modules via events.

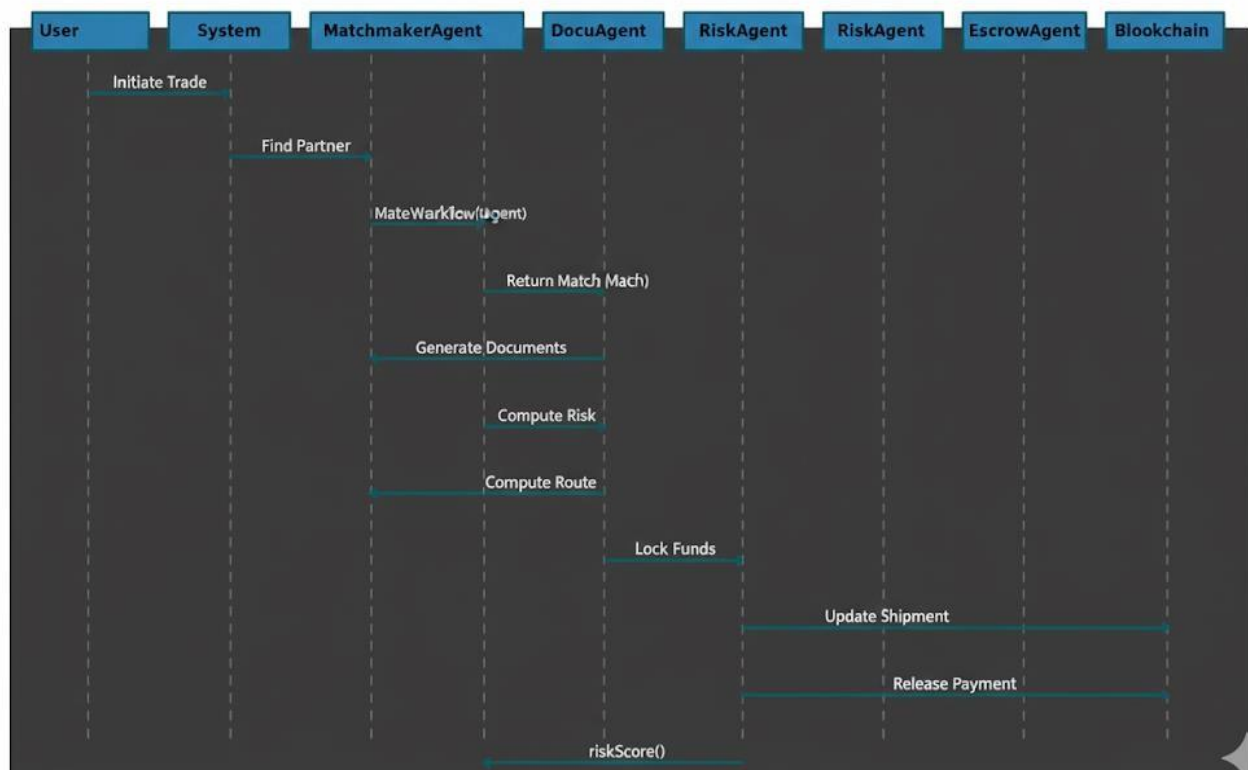
5.3.3 System Workflow (High-Level)

SME registers and is verified.

1. MatchmakerAgent proposes trade partners.
2. SMEs confirm trade → workflow starts.
3. DocuAgent generates required documents.
4. RiskAgent computes risk profile.
5. LogisticsAgent plans shipment route.
6. Shipment is executed.
7. EscrowAgent confirms conditions and releases payment.

5.3.4 Sequence Diagram (End-to-End Trade Execution)

Trade Transaction Flow



1. **User** → **System**: Initiate trade
2. **System** → **MatchmakerAgent**: Find partner
3. **MatchmakerAgent** → **System**: Return match
4. **System** → **DocuAgent**: Generate documents
5. **System** → **RiskAgent**: Compute risk
6. **System** → **LogisticsAgent**: Compute route
7. **System** → **EscrowAgent**: Lock funds
8. **LogisticsAgent** → **System**: Update shipment
9. **EscrowAgent** → **Blockchain**: Release payment

5.4.2 Authentication Model

5.4.2.1 **OAuth2 + JWT** for user-facing APIs

5.4.2.2 Internal agents use **service keys** for inter-agent communication

5.4.2.3 Tokens contain:

5.4.2.3.1 Role

5.4.2.3.2 Permissions

5.4.2.3.3 Expiry

5.4.2.3.4 Risk constraints

5.4.3 Agent-to-Agent Communication

Messaging Bus: **RabbitMQ / Kafka**

Protocol: **Event-driven, asynchronous JSON messages**

Common message topics:

5.4.3.1 `documents.ready`

5.4.3.2 `risk.updated`

5.4.3.3 `logistics.route_planned`

5.4.3.4 `escrow.release_triggered`

5.4.4 Event-Driven Workflow

Example event chain:

```
verification_complete →  
MatchmakerAgent deal_confirmed →  
DocuAgent docs_generated → RiskAgent  
risk_score_ready → LogisticsAgent  
delivery_confirmed → EscrowAgent
```

Chapter 6: Technology Stack

This chapter describes the implementation strategy adopted for developing the TradeOS platform. It details the technology stack, module-wise development, agent implementation, system integration, and the deployment environment used for validating the prototype.

6.1 Technology Stack

The implementation of TradeOS uses a combination of modern web technologies, AI frameworks, and distributed ledger components to support agent-based automation and secure trade execution.

Component	Technology / Tools	Purpose / Description
Backend	Python 3.10+	Core language for business logic and autonomous agents
	FastAPI	High-performance backend framework for REST APIs
	Uvicorn / Gunicorn	ASGI servers for running and deploying FastAPI applications
Agent Framework	LangChain / Autogen	Supports autonomous agent workflows, reasoning, tool-calling, and inter-agent communication
AI & NLP	HuggingFace Transformers	HS Code classification and document understanding
	Tesseract OCR	Extracts text from invoices and packing lists
	NumPy, Pandas, Scikit-learn	Data preprocessing, feature engineering, and statistical modelling
	Custom Stochastic Models	Risk estimation and predictive analytics
Frontend	React.js	User interface for marketplace, dashboards, and trade tracking
	Axios	Used for frontend–backend API communication
Trust Layer / Blockchain	Solidity (Smart Contracts)	Used by EscrowAgent for secure, milestone-based payments
	Web3.py	Connects backend services to the blockchain network
Database	PostgreSQL	Persistent relational storage for users, trades, documents, and logs
	SQLAlchemy ORM	Object-Relational Mapping between Python models and database tables
Deployment	Docker	Containerization for consistent and portable deployments
	Cloud-Ready Design	Supports deployment on AWS, Azure, or GCP infrastructure

Chapter 7: Implementation

7.1 Agent Implementations

Agent	Functions	Key Logic
MatchmakerAgent	Identifies compatible buyers/sellers	Vector-based matching
	Uses similarity scoring, product matching, and demand–supply metrics	Filter constraints (location, product category, certifications)
	Returns top-ranked trade partners	
DocuAgent	Extracts text from invoices and packing lists using OCR	BERT/NMT-based classification
	Performs HS Code classification using Transformer-NLP models	Template-based document synthesis
	Generates customs documentation (BL, COO, Invoice Draft)	
RiskAgent	Collects external data (weather, port delays, currency volatility)	Monte Carlo simulation
	Computes risk scores using stochastic modelling	Weighted risk scoring function
	Supplies dynamic risk constraints to LogisticsAgent	
LogisticsAgent	Integrates with shipping carrier APIs	Graph-based routing algorithms (Dijkstra/A*)
	Computes optimal routes based on risk, cost, and delivery time	Risk-adjusted route scoring
	Provides a logistics plan and estimated timelines	
EscrowAgent	Deploys smart contracts for secure payment holding	Solidity-based smart contract conditions
	Releases funds only when delivery confirmation is received	Blockchain event listening
	Maintains immutable transaction records on blockchain	

7.2 Frontend / Backend Integration

A seamless integration strategy ensures real-time updates and dynamic workflow execution:

7.2.1 **React frontend** communicates with the FastAPI backend via REST APIs

7.2.2 Agent status updates are pushed to the UI using **WebSockets**

7.2.3 Backend triggers agent workflows asynchronously

7.2.4 Smart contract state updates are reflected on UI dashboards via Web3.py listeners

Data Flow:

User → React UI → FastAPI → Agents → Database → Blockchain → UI updates

7.3 Deployment Environment

Local Environment

7.3.1 Dockerized containers for backend, database, and agents

7.3.2 Local blockchain test network (Ganache/Hardhat) for EscrowAgent testing

Cloud-ready Architecture

The system is designed for deployment on cloud platforms such as:

7.3.3 AWS (EC2 + RDS + ECR)

7.3.4 Azure (VM + Azure SQL)

7.4 Project Execution and Management Plan

Software Methodology: Agile Scrum

The development of TradeOS follows an **Agile Scrum methodology** over four sprints (12 weeks). This ensures iterative development, continuous feedback, and incremental delivery of working software components.

Key characteristics of Scrum applied in this project:

7.4.1 **Time-boxed sprints** of two weeks each

7.4.2 **Daily stand-up meetings** to track progress and remove blockers

7.4.3 **Incremental delivery** of functional modules at the end of each sprint

7.4.4 **Sprint reviews and retrospectives** to refine system requirements and technical strategy

7.4.5 **Backlog grooming** to prioritize features based on feasibility and dependency

Sprint Plan Overview

Sprint	Goal / Theme	Key Deliverables	Technical Focus
S1: Foundation & Verification	Build marketplace UI and onboarding workflow	Basic UI, backend setup, verification module	Infrastructure setup, data handling
S2: Core Automation Logic	Implement partner discovery and documentation automation	MatchmakerAgent, initial DocuAgent (HS Code classification)	NLP models, matching algorithms
S3: Intelligence & Risk Mitigation	Introduce intelligent decision-making agents	RiskAgent, integrated LogisticsAgent	Stochastic modelling, logistics API
S4: Final Automation & Deployment	Deploy full system and automate end-to-end trade execution	EscrowAgent, complete trade simulation	Smart contracts, system integration

This methodology ensures modular development, enabling each agent to be designed, validated, and integrated independently.

7.5 Risk Assessment and Mitigation

Risk Category	Key Risk	Mitigation Strategy
Technical Risk	HS Code classification may not achieve required accuracy	Use robust open-source Transformer models; adjust model size for optimal inference speed; implement fallback rule-based classification
Integration Risk	Difficulty integrating LogisticsAgent with external APIs or blockchain	Build API mocks in Sprint 1; follow standard REST structures; isolate DLT calls in service modules
Timeline Risk	Delays in complex agent logic (M3/M4)	M1 ensures daily monitoring; pair-programming on complex modules; strict adherence to sprint commitments

Chapter 8: Result And Analysis

To empirically validate the effectiveness of the TradeOS multi-agent automation framework, a structured, evidence-based evaluation methodology was followed. The goal of this validation is to demonstrate that the coordinated agentic workflow of TradeOS provides measurable improvements over conventional manual or rule-based trade processes. All tests were executed under controlled, repeatable conditions to ensure scientific reliability.

8.1 Baseline for Comparison: The “Manual Workflow Model”

To establish a meaningful benchmark, a baseline **Manual Workflow Model** was defined. This model represents the traditional SME trade process, characterised by:

- 8.1.1 Manual partner identification
- 8.1.2 Manual document checking and HS classification
- 8.1.3 No predictive risk modelling
- 8.1.4 Static logistics routing based solely on least cost
- 8.1.5 Manual bank transfers without contract-backed safeguards

This baseline captures the existing industry challenges and serves as the **control model** against which TradeOS is evaluated. Comparing TradeOS to this baseline enables quantifiable proof of automation, speed, and accuracy improvements introduced by the agentic architecture.

8.2 Trade Simulation Environment

A controlled simulation environment was created to enable repeatable and realistic validation of the system.

- A. Trade Route Scenario:** A standardized Bengaluru → Hamburg route was used, incorporating multiple shipping paths, weather variability, port delays, and customs requirements.
- B. Document Dataset:** Synthetic, industry-formatted invoices and packing lists were prepared to test OCR accuracy, HS Code classification, and document generation.
- C. Logistics & Risk Data Simulator:** A Python simulator produced dynamic inputs such as weather changes, port congestion levels, and currency volatility, enabling controlled testing of the RiskAgent and LogisticsAgent.

8.3 Key Performance Indicators (KPIs)

The effectiveness is validated by comparing system outputs against the baseline model using the following KPIs:

KPI	Description	Expected Outcome
1. Documentation Processing Accuracy	Measures correctness of HS Code classification and extraction of invoice fields.	Significant improvement compared to manual interpretation.
2. Trade Execution Time	Total time from partner discovery to final settlement.	Reduced workflow duration due to parallel multi-agent execution.
3. Route Optimization Efficiency	Evaluates difference between system-selected route and optimal cost-risk-balanced route.	Lower risk-adjusted delivery time.
4. Risk Prediction Consistency	Assesses stability of stochastic risk scores across repeated simulations.	Low variance indicating robust modelling.
5. Settlement Security & Transparency	Validates correctness and immutability of smart contracts and milestone-based fund release.	Secure, tamper-proof, and automated settlement.

8.4 Outcome Interpretation

Validation success is achieved if TradeOS consistently demonstrates:

8.4.1 **Higher documentation accuracy** than the manual baseline

8.4.2 **Reduced trade execution time** through parallel agent coordination

8.4.3 **More resilient route decisions** through risk-aware optimisation

8.4.4 **Secure, tamper-proof payments** via blockchain escrow

8.4.5 **Improved transparency and traceability** across all trade events

These results collectively confirm the value of adopting a coordinated multi-agent system for SME cross-border trade automation.

Chapter 8: Conclusion

The proposed TradeOS platform demonstrates the effective integration of multiple autonomous AI agents to streamline and optimize the end-to-end international trade process. By leveraging structured trade datasets such as UN Comtrade along with real-time data sources like OpenWeather and ExchangeRate-API, the system transitions from static decision-making to a dynamic, data-driven framework.

The architecture combines specialized agents—Matchmaker, Negotiation, Risk Assessment, Document Generation, and Logistics—each responsible for a critical stage in the trade lifecycle. This modular approach ensures scalability, flexibility, and robustness while maintaining transparency in decision-making. The Risk Agent enhances reliability by incorporating multi-dimensional risk factors, while the Logistics Agent optimizes transportation planning through intelligent route and mode selection using platforms such as OpenStreetMap.

Furthermore, the system successfully reduces manual intervention by automating complex processes such as supplier selection, price negotiation, compliance documentation, and shipment planning. Although certain elements rely on synthetic or semi-realistic datasets due to limitations in accessing real-world proprietary trade data, the overall framework closely simulates real-world trade operations and provides a strong foundation for future deployment.

In conclusion, TradeOS represents a significant step toward intelligent, autonomous trade ecosystems, offering improved efficiency, reduced risk, and enhanced decision support. Future enhancements may include deeper integration of real-time APIs, machine learning-based predictive models, and direct connectivity with financial and logistics service providers to further increase system accuracy and real-world applicability.

APPENDIX – Publication Details

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